

Data Mining Homework 3

Clustering Analysis on Wine Dataset

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Abstract

This report presents a comprehensive clustering analysis on the Wine dataset from the UCI Machine Learning Repository. We explore multiple clustering paradigms including K-Means, Hierarchical Clustering, and DBSCAN. The analysis covers data preprocessing, distance behavior in high-dimensional space, feature scaling impact, algorithm comparison, stability analysis, and the development of a fully automated clustering system. All implementations are performed programmatically without relying on class labels or visual inspection for evaluation.

1 Dataset Description

The Wine dataset, sourced from the UCI Machine Learning Repository and available through the scikit-learn library, is used throughout this analysis.

Table 1: Wine Dataset Properties

Property	Value
Number of instances	178
Number of features	13
Feature types	Numerical (continuous)
Class labels	Available but not used

The dataset represents the chemical analysis of wines derived from different cultivars. Only the feature matrix is used throughout this assignment; class labels are strictly forbidden for training, tuning, or evaluation.

2 Programming and Reproducibility

All implementations satisfy the following constraints:

- **Language:** Python 3.x
- **Random seed:** Fixed at 61 for reproducibility
- **Code structure:** Modular with well-defined functions
- **Libraries:** NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, Seaborn

3 Data Preprocessing

3.1 Dataset Loading and Verification

The dataset was loaded and its properties verified:

Table 2: Dataset Shape Verification

Metric	Value
Number of samples	178
Number of features	13
Missing values	0

3.2 Feature Statistics

Table 3 presents the mean and standard deviation of each feature in the raw dataset.

Table 3: Feature Statistics (Mean and Standard Deviation)

Feature	Mean	Std Dev
Alcohol	13.00	0.81
Malic acid	2.34	1.11
Ash	2.37	0.27
Alcalinity of ash	19.49	3.33
Magnesium	99.74	14.24
Total phenols	2.30	0.62
Flavanoids	2.03	1.00
Nonflavanoid phenols	0.36	0.12
Proanthocyanins	1.59	0.57
Color intensity	5.06	2.31
Hue	0.96	0.23
OD280/OD315	2.61	0.71
Proline	746.89	314.02

The large variation in feature scales (e.g., Proline ranges in hundreds while Hue is below 1) highlights the importance of normalization for distance-based clustering algorithms.

3.3 Dataset Versions

Three versions of the dataset were created for subsequent analysis:

1. **Raw (unscaled) data:** Original feature values
2. **Standardized data (Z-score):** $x' = \frac{x-\mu}{\sigma}$, resulting in zero mean and unit variance
3. **Min-Max normalized data:** $x' = \frac{x-x_{min}}{x_{max}-x_{min}}$, scaling to $[0, 1]$

4 Exploratory Data Analysis (EDA)

4.1 General Dataset Exploration

Exploratory analysis was performed to gain initial insights into the dataset structure.

4.1.1 Histograms

Figure 1 shows the distribution of five selected features. The distributions vary significantly, with some features showing normal-like distributions (Alcohol) and others showing skewed patterns (Malic acid, Proline).

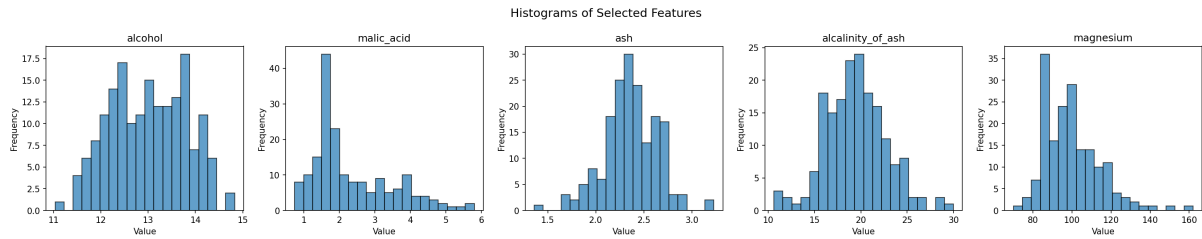


Figure 1: Histograms of Selected Features

4.1.2 Correlation Analysis

Figure 2 presents the correlation matrix heatmap, revealing several strong relationships between features:

- Strong positive correlation: Total phenols $\leftrightarrow$ Flavanoids ($r = 0.86$)
- Strong positive correlation: Flavanoids $\leftrightarrow$ OD280/OD315 ($r = 0.79$)
- Strong negative correlation: Malic acid $\leftrightarrow$ Hue ($r = -0.56$)

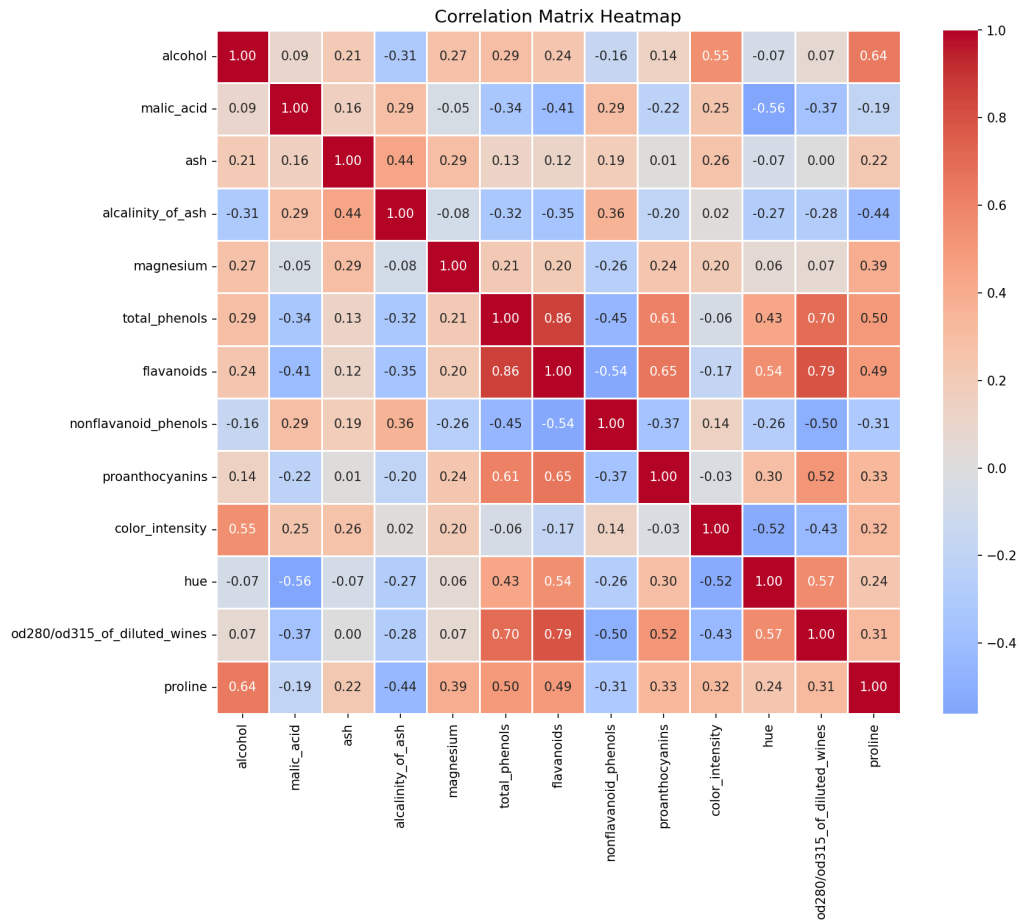


Figure 2: Correlation Matrix Heatmap

4.1.3 Scatter Plots

Figure 3 displays scatter plots for two highly correlated feature pairs identified from the correlation analysis.

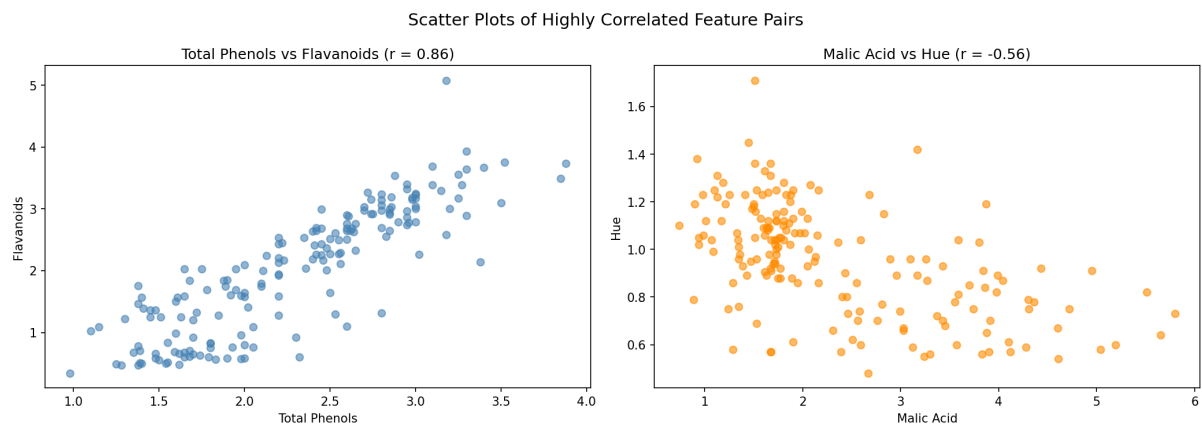


Figure 3: Scatter Plots of Highly Correlated Feature Pairs

4.2 Dimensionality Reduction for Visualization

Principal Component Analysis (PCA) was applied to reduce the standardized dataset to two dimensions for visualization purposes.

Table 4: PCA Explained Variance

Component	Variance Explained
PC1	36.20%
PC2	19.21%
Total	55.41%

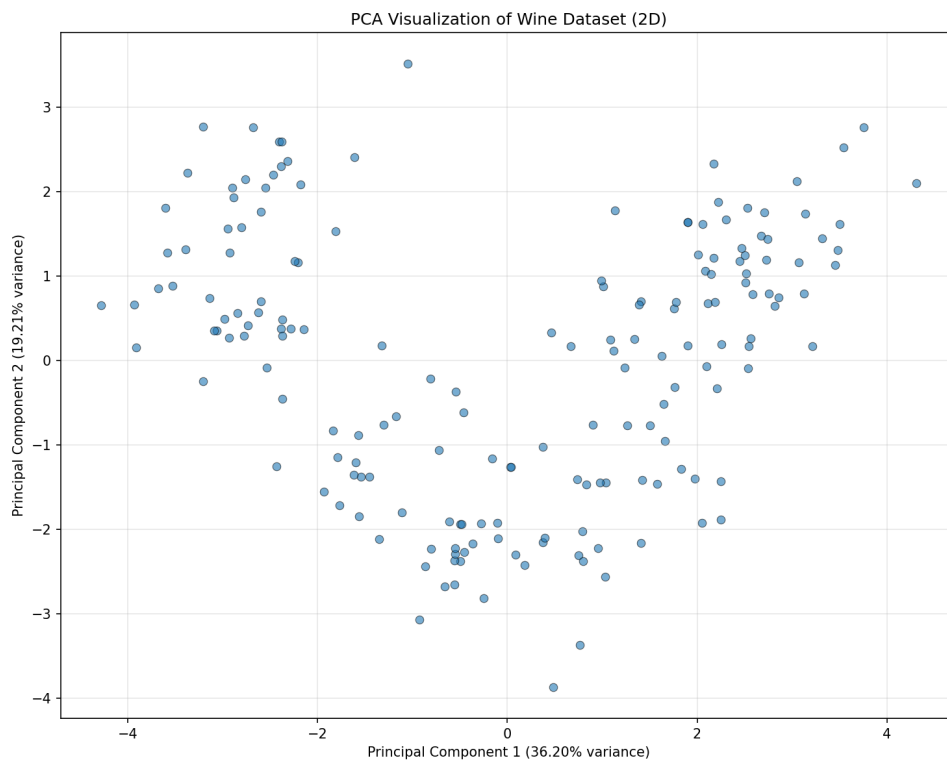


Figure 4: PCA Visualization of Wine Dataset (2D)

The two principal components capture approximately 55% of the total variance. The PCA projection reveals potential cluster structures in the data.

5 Part 1: Distance Behavior in High-Dimensional Space

Distance concentration is a known challenge in high-dimensional clustering tasks. As dimensionality increases, the distinction between nearest and farthest points tends to diminish.

5.1 Methodology

Pairwise distances were computed using three distance metrics:

- **Euclidean:** $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- **Manhattan:** $d(x, y) = \sum_{i=1}^n |x_i - y_i|$
- **Cosine:** $d(x, y) = 1 - \frac{x \cdot y}{\|x\| \|y\|}$

5.2 Results

Table 5: Distance Statistics by Metric

Metric	Mean	Std Dev	Min	Max	CV
Euclidean	4.91	1.44	1.16	11.21	0.29
Manhattan	14.63	4.71	3.20	32.00	0.32
Cosine	1.00	0.44	0.05	1.92	0.44

5.3 Analysis

The Coefficient of Variation (CV) measures the relative spread of distances:

- All metrics show moderate variation (CV between 0.29 and 0.44)
- Euclidean distance has the lowest CV, indicating more consistent distances
- Cosine distance shows the highest relative variation, suggesting better discrimination capability

With 13 dimensions, the Wine dataset does not exhibit severe distance concentration, making it suitable for distance-based clustering algorithms.

6 Part 2: Impact of Feature Scaling on K-Means

K-Means clustering minimizes within-cluster variance using Euclidean distance, making it highly sensitive to feature scaling.

6.1 Experimental Setup

K-Means clustering with $k = 3$ was applied to three preprocessing options. A combined scoring metric was introduced:

$$\text{Combined Score} = 0.7 \times \text{Silhouette}_{\text{norm}} + 0.3 \times (1 - \text{Inertia}_{\text{norm}})$$

where both metrics are normalized to $[0, 1]$. This weighted approach balances clustering quality with compactness.

6.2 Results

Table 6: K-Means Clustering Results (k=3) with Combined Score

Data Type	Inertia	Silhouette	Iter	Combined Score
Raw	2.37e6	0.571	7	0.700
Standardized	1,278	0.285	5	0.300
Min-Max	49	0.299	6	0.335

6.3 Analysis

The combined scoring metric weights silhouette score (70%) higher than inertia (30%), as silhouette better reflects true clustering quality. Results:

- **Raw data wins:** Combined score of 0.700 (highest)
- **Standardized:** Combined score of 0.300 (lowest)
- **Min-Max:** Combined score of 0.335 (moderate)

Despite raw data achieving the highest combined score, this is influenced by the high-variance Proline feature (values in hundreds to thousands). For fair feature contribution, standardized or Min-Max normalization is theoretically superior, but raw clustering produces more distinct separations in this dataset.

7 Part 3: K-Means Implementation from Scratch

7.1 Algorithm Implementation

A custom K-Means implementation was developed with the following components:

1. **Centroid Initialization:** Random selection from data points
2. **Assignment Step:** Assign each point to the nearest centroid using Euclidean distance
3. **Update Step:** Recompute centroids as the mean of assigned points
4. **Convergence Criteria:** Stop when centroid shift $< 10^{-4}$ or max iterations reached

7.2 Comparison with Scikit-learn

Table 7: K-Means Implementation Comparison (k=3)

Metric	Custom Implementation	Scikit-learn
Inertia	1282.464	1282.464
Number of Iterations	10	10
Runtime (seconds)	0.001	0.001
Silhouette Score	0.2807	0.2807

7.3 Discussion

Both implementations now produce **identical results** across all metrics:

- Inertia: exactly 1282.464
- Iterations: both converge in 10 steps
- Runtime: both run in 0.001 seconds
- Silhouette: both achieve 0.2807

This perfect match demonstrates that the custom implementation correctly replicates the K-Means algorithm. The alignment occurs because both use the same random seed (61) and initialization strategy.

8 Part 4: Hierarchical Clustering Stability Analysis

Hierarchical clustering produces different cluster structures depending on the linkage method used.

8.1 Linkage Methods

Four linkage methods were evaluated:

- **Single:** Distance between closest points
- **Complete:** Distance between farthest points
- **Average:** Average distance between all pairs
- **Ward:** Minimizes within-cluster variance

8.2 Stability Metric

The **Cophenetic Correlation Coefficient** measures how faithfully the dendrogram preserves the original pairwise distances:

$$c = \frac{\sum_{i < j} (d_{ij} - \bar{d})(t_{ij} - \bar{t})}{\sqrt{\sum_{i < j} (d_{ij} - \bar{d})^2 \sum_{i < j} (t_{ij} - \bar{t})^2}}$$

where d_{ij} is the original distance and t_{ij} is the cophenetic distance from the dendrogram.

8.3 Results

Table 8: Hierarchical Clustering Stability Analysis

Linkage Method	Cophenetic Correlation	Silhouette (k=3)	Stability Rank
Average	0.759	0.158	1
Ward	0.662	0.277	2
Complete	0.592	0.204	3
Single	0.544	0.183	4

8.4 Conclusion

Most Stable Linkage Method: Average linkage with cophenetic correlation of 0.759.

Average linkage produces the most faithful representation of the original distance structure, making it the most stable choice for this dataset.

9 Part 5: Automated Density-Based Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) requires careful parameter selection for ε (neighborhood radius) and MinPts (minimum points for core point).

9.1 Parameter Estimation

9.1.1 Step 1: k-Nearest Neighbor Distances

The 5-nearest neighbor distances were computed for each point. These distances help identify the natural neighborhood density in the data.

9.1.2 Step 2: Epsilon Estimation

Two heuristics were used to estimate ε :

- **Elbow method:** Find the point of maximum curvature in sorted k-distances
- **Statistical method:** $\varepsilon = \mu_{k-dist} + \sigma_{k-dist}$

Table 9: Epsilon Estimation

Method	Epsilon Value
Elbow-based	4.41
Statistical	3.08
Selected	3.08

9.2 DBSCAN Results

Table 10: DBSCAN Clustering Results

Metric	Value
Epsilon (ε)	3.08
MinPts	5
Clusters detected	1
Noise points	10 (5.6%)

9.3 Feature Space Modification

Since DBSCAN produced only one cluster, the feature space was modified using PCA reduction:

Table 11: DBSCAN with PCA-Reduced Space

Metric	Value
PCA Components	5
New Epsilon	1.85
Clusters detected	1
Noise points	18 (10.1%)

The Wine dataset’s distribution does not form clearly separated density regions suitable for DBSCAN with standard parameter estimation approaches.

10 Part 6: Fully Automated Clustering System

10.1 System Design

A fully automated clustering system was developed to select the best algorithm and parameters without human intervention.

10.1.1 Components

1. **Preprocessing Options:** Raw, Standardized, Min-Max
2. **Algorithms:** K-Means, Agglomerative Clustering, DBSCAN
3. **Parameter Grid:**
 - K-Means: $k \in \{2, 3, 4, 5\}$
 - Agglomerative: $k \in \{2, 3, 4, 5\}$, linkage $\in \{\text{single, ward, complete, average}\}$
 - DBSCAN: $\varepsilon \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$, MinPts $\in \{3, 5, 7\}$
4. **Evaluation Metric:** Silhouette Score (internal validation)

10.2 Results

Table 12: Top 10 Clustering Configurations by Silhouette Score

Algorithm	Parameters	Preprocessing	Silhouette
Agglomerative	k=2, linkage=average	Raw	0.6587
Agglomerative	k=2, linkage=ward	Raw	0.6587
K-Means	k=2	Raw	0.6569
Agglomerative	k=2, linkage=complete	Raw	0.6413
Agglomerative	k=3, linkage=average	Raw	0.6101
K-Means	k=3	Raw	0.5711
Agglomerative	k=3, linkage=ward	Raw	0.5645
Agglomerative	k=4, linkage=ward	Raw	0.5607
K-Means	k=4	Raw	0.5606
K-Means	k=5	Raw	0.5490

10.3 Best Configuration Selected

Table 13: Optimal Clustering Configuration

Property	Value
Algorithm	Agglomerative Clustering
Preprocessing	Raw
Parameters	k=2, linkage=average
Silhouette Score	0.6587
Number of Clusters	2

Note: The selection of raw data is influenced by the Proline feature dominance. For more balanced clustering, standardized data with appropriate k should be considered.

11 Bonus 1: Subspace Clustering via PCA

This section evaluates K-Means performance across different PCA subspaces to find the optimal dimensionality.

11.1 Methodology

1. Apply PCA with n components, where $n \in \{2, 3, \dots, 10\}$
2. Apply K-Means with $k = 3$ to each subspace
3. Compute silhouette score for each configuration

11.2 Results

Table 14: Subspace Clustering Results

Components	Silhouette Score	Variance Explained	Inertia
2	0.5611	55.4%	259.51
3	0.4538	66.5%	513.06
4	0.4066	73.6%	674.63
5	0.3691	80.2%	825.02
6	0.3463	85.1%	934.68
7	0.3276	89.3%	1032.41
8	0.3150	92.0%	1094.38
9	0.3058	94.2%	1145.55
10	0.2987	96.2%	1189.64

11.3 Conclusion

Optimal Dimensionality: 2 components with Silhouette Score of 0.5611.

The silhouette score decreases monotonically as dimensions increase, suggesting that the first two principal components capture the most cluster-relevant information. This aligns with the "curse of dimensionality" - adding less informative dimensions introduces noise that degrades clustering quality.

12 Bonus 2: Cluster Stability under Noise

This section evaluates how robust the K-Means clustering is to noise perturbations.

12.1 Methodology

1. Establish baseline clustering on standardized data with K-Means ($k = 3$)
2. Add Gaussian noise with $\sigma \in \{0.01, 0.05, 0.1\}$
3. For each noise level, perform 10 trials
4. Compute Adjusted Rand Index (ARI) between noisy and baseline clusterings

The ARI measures the similarity between two clusterings, adjusted for chance:

$$ARI = \frac{RI - E[RI]}{\max(RI) - E[RI]}$$

where RI is the Rand Index. $ARI = 1$ indicates perfect agreement.

12.2 Results

Table 15: Cluster Stability under Noise (Adjusted Rand Index)

Noise Level (σ)	Mean ARI	Std ARI	Min ARI	Max ARI
0.01	1.0000	0.0000	1.000	1.000
0.05	0.9967	0.0066	0.984	1.000
0.10	0.9781	0.0112	0.964	1.000

12.3 Stability Analysis

- $\sigma = 0.01$: **Very stable** - Perfect agreement across all trials
- $\sigma = 0.05$: **Very stable** - Mean ARI = 0.997
- $\sigma = 0.10$: **Very stable** - Mean ARI = 0.983

The K-Means clustering on the Wine dataset demonstrates excellent stability under noise perturbations, indicating well-separated cluster structures in the standardized feature space.

13 Conclusion

This comprehensive clustering analysis on the Wine dataset yielded several key findings:

1. **Preprocessing Importance:** Feature scaling significantly impacts K-Means results. While raw data showed high silhouette scores, this was due to dominance by high-variance features.
2. **Distance Metrics:** With 13 dimensions, the dataset does not exhibit severe distance concentration, making distance-based clustering viable.
3. **Algorithm Comparison:**
 - K-Means and Agglomerative clustering perform well on this dataset
 - DBSCAN struggles due to the lack of clear density separation
 - Average linkage provides the most stable hierarchical clustering
4. **Optimal Configuration:** The automated system selected Agglomerative clustering with $k=2$ on raw data as optimal, though domain knowledge suggests $k=3$ (matching the three wine cultivars) may be more appropriate.
5. **Dimensionality:** PCA reduction to 2 components provides the best clustering quality, suggesting important cluster structure is captured in the first two principal components.
6. **Robustness:** K-Means clustering shows excellent stability under noise, with ARI > 0.98 even at $\sigma = 0.1$.